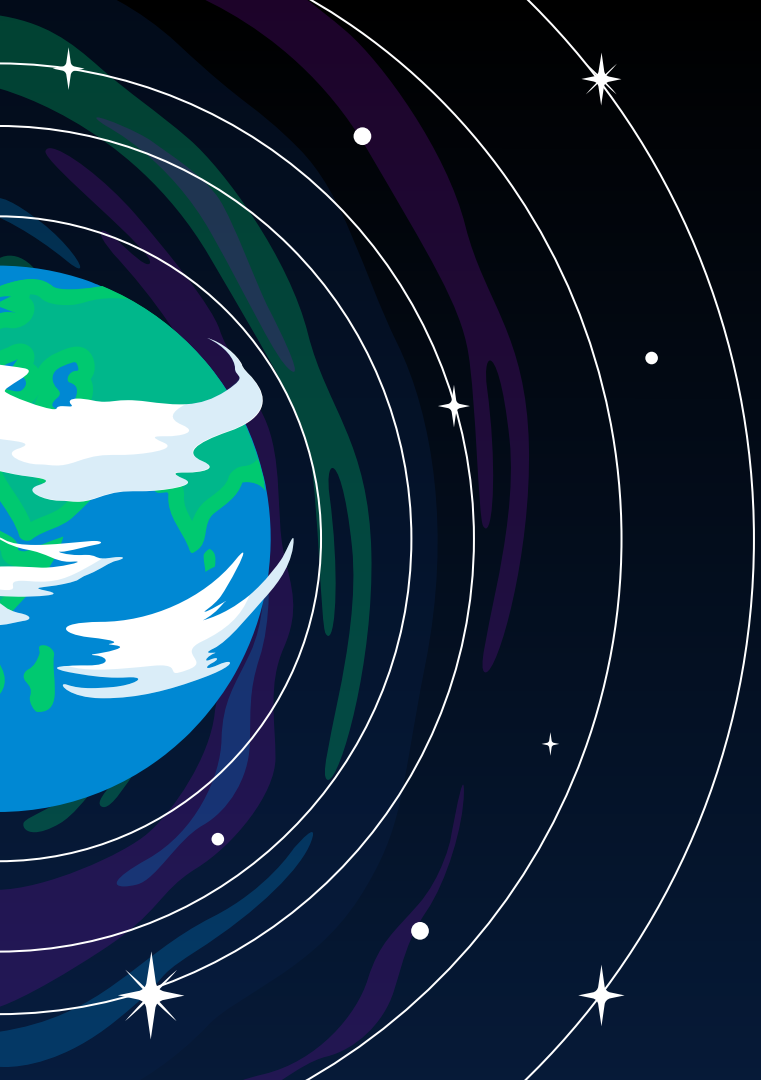




Rocket Reusability

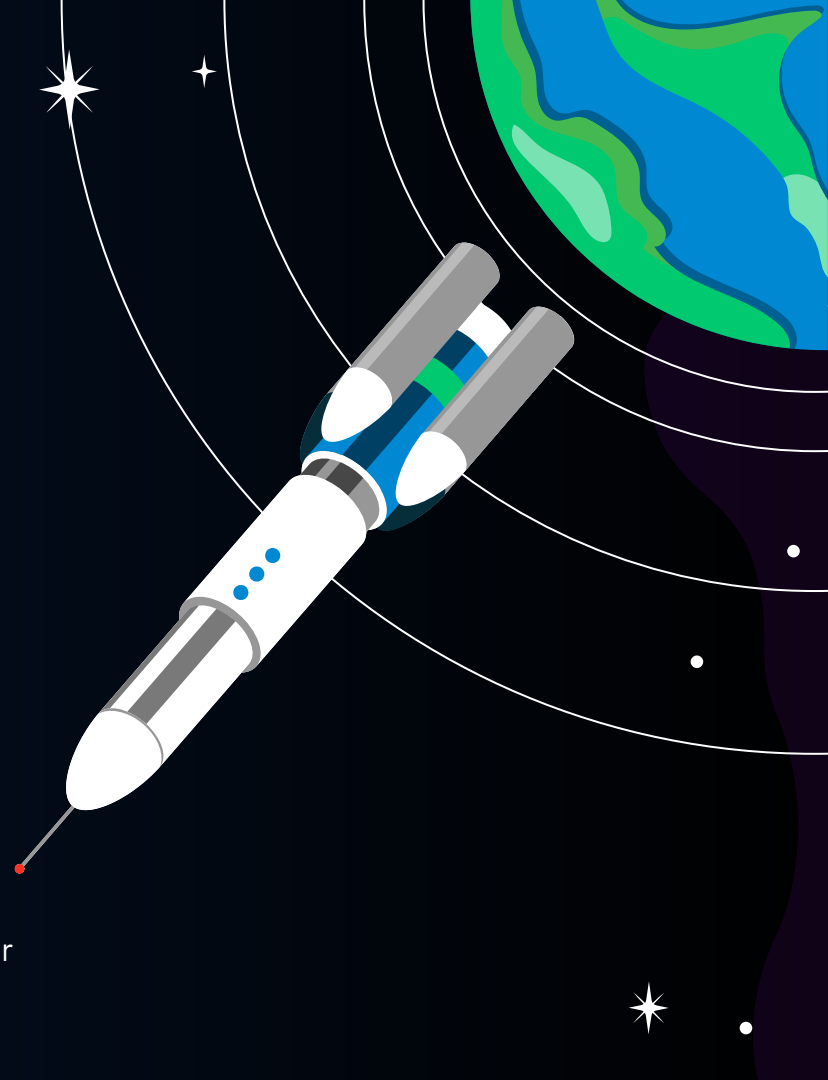
Lakshya Kolli

Background - Why rocket reusability matters

- Space exploration used to be very wasteful, with early rockets being solely single use
- Private companies like SpaceX started experimenting and testing reusable launch vehicles (RLVs) to make space exploration cheaper and cleaner
- This topic interests me because I am interested in space exploration, and with the help of rocket reusability, space exploration will become more accessible and sustainable for future generations

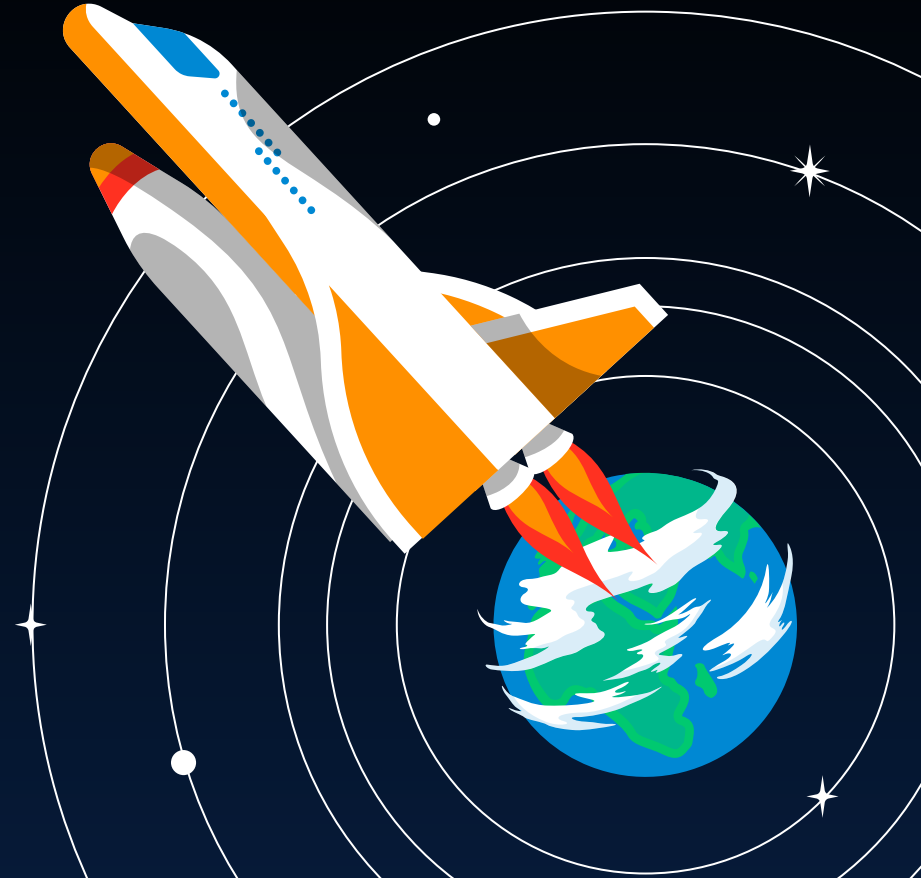
Why this topic matters

- Major setbacks to space travel: High Cost and Environmental Impact
- The development of reusable rockets solve both issues
- Economically, the cost of each mission is lowered, and environmentally, waste is minimized
- Opens the door for more frequent exploration in space
- Shows that space progress could be made without harming our planet



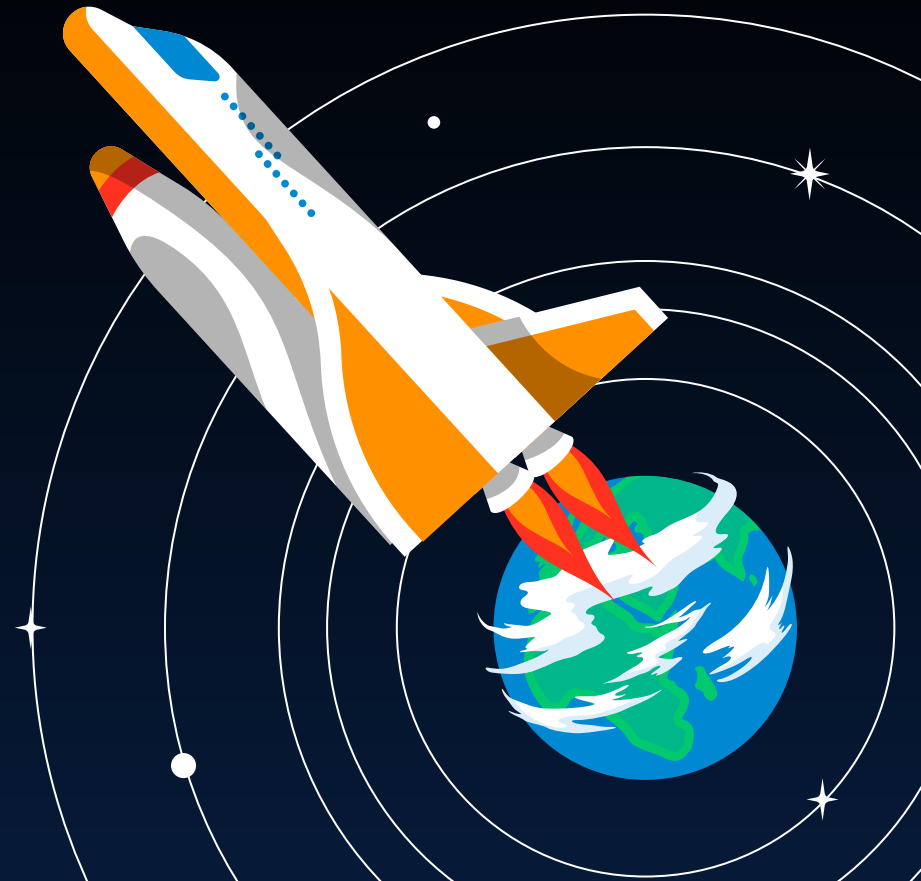
Research question:

Research question: In what ways can advancements in rocket reusability lower the long-term costs of space exploration and make future missions more sustainable?



Hypothesis:

Hypothesis: The overall cost and sustainability of space exploration will be significantly improved by advancements in rocket reusability through the reduction of manufacturing expenses, decreased need for new materials, and increased frequency of launches.



Rationale and basis

- Rocket reusability addresses the high cost and environmental impact of space exploration
- Waste is reduced as companies spend less capital on building new parts and use fewer materials
- Less debris and waste left after launches
- Research shows that the total lifetime cost of reusable rockets are reduced if they are designed to be reused many times
- Quick repairs and fast completion times make spaceflight more efficient
- Pollution and resource use that come with building new rockets are reduced



Future plans

- Explore more NASA articles and academic sources that are related to rocket reusability
- Compare different studies to see how rocket reusability affects short term and long term costs
- Focus on quantitative data to like launch frequency, refurbishment time, and number of reuses to find trends from different studies

THANKS!



Does anyone have any questions?

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